Experiment-9: Analysis of revenue in sales dataset:

i) Create a choropleth map (fill the map) to spot the special trends to show the state which has the highest revenue.

ii) Create a line chart to show the revenue based on the month of the year.

iii) Create a bin of size 10 for the age measure to create a new dimension to show the revenue.

iv) Create a donut chart view to show the percentage of revenue per region by creating zero access in the calculated field using powerbi

v) Create a butterfly chart by reversing the bar chart to compare female & male revenue based on product category.

vi) Create a calculated field to show the average revenue per state & display profitable & non-profitable state. vii) Build a dashboard.

Step 1: Import the Sales Dataset

1. Open Power BI Desktop.
2. Click **Get Data** > **Excel** or **Text/CSV**, depending on your dataset format.
3. Select the file containing your sales data and click **Load**.

i) Choropleth Map to Spot the State with Highest Revenue

1. **Add a Map Visualization**:
   - Select the **Filled Map** visualization.
2. **Configure the Map**:
   - Drag State to the **Location** field.
   - Drag Revenue to the **Values** field.
3. **Customize the Map**:
   - Use conditional formatting to highlight states with the highest revenue.
   - Add tooltips to display detailed information.

ii) Line Chart to Show Revenue by Month

1. **Add a Line Chart**:
   - Select the **Line Chart** visualization.
2. **Configure the Chart**:
   - Drag Month (ensure it's in Date or Text format) to the **Axis** field.
   - Drag Revenue to the **Values** field.
3. **Format the Chart**:
   - Sort the months chronologically.
   - Add data labels to display revenue values.

iii) Create a Bin for Age Measure

1. **Create Age Bins**:
   - Go to the **Modeling** tab and click **New Group**.
   - Select Age as the field, set **Bin Size** to 10, and name the group (e.g., Age Bin).
2. **Visualize Age Bins**:
   - Use a bar chart or column chart:
     - Drag Age Bin to the **Axis** field.
     - Drag Revenue to the **Values** field.

iv) Donut Chart for Revenue Percentage Per Region

1. **Create Zero Access for Calculated Field**:
   - Go to the **Modeling** tab and click **New Measure**.
   - Create a measure for zero access: Zero Access = 0.
2. **Add a Donut Chart**:
   - Drag Region to the **Legend** field.
   - Drag Revenue to the **Values** field.
3. **Customize the Chart**:
   - Enable percentages in the data labels.
   - Format the chart to clearly display regions and their respective revenue.

v) Butterfly Chart for Gender-Based Revenue by Product Category

1. **Prepare the Data**:
   - Ensure Gender, Product Category, and Revenue fields are available.
2. **Create a Bar Chart**:
   - Add Product Category to the **Axis** field.
   - Add Revenue to the **Values** field and filter by **Male**.
3. **Reverse the Bar Chart**:
   - Copy the chart and modify it for **Female** revenue.
   - Reverse the axis for one chart.
4. **Combine Charts**:
   - Align the two charts side by side to create the butterfly effect.

vi) Calculate Average Revenue per State and Identify Profitable States

1. **Create a Calculated Field**:
   - Go to the **Modeling** tab and create a new measure:

     ```
     AvgRevenuePerState = AVERAGE(Sales[Revenue])
     ```

   - Create another measure to classify profitability:

     ```
     Profitability = IF(SUM(Sales[Revenue]) > AvgRevenuePerState, "Profitable", "Non-Profitable")
     ```

2. **Visualize**:
   - Use a map or table to display states as profitable or non-profitable.

vii) Build a Dashboard